

# Complete Source Code Tasks 1–5

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**Course:** Web Technologies

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# Task 1 — World Wide Web Concepts

Folder: Assignment-WebTech-2026-08-18/Task1-WWW-Concepts/

● index.html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Understanding the World Wide Web</title>
7  <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10
11  <!-- Page header: introduces the topic to the reader -->
12  <header class="page-header">
13    <h1>Understanding the World Wide Web</h1>
14    <p class="subtitle">A quick reference built for a new institution's website team</p>
15  </header>
16
17  <!-- Simple in-page navigation, one link per concept section -->
18  <nav class="toc">
19    <a href="#internet">Internet</a>
20    <a href="#network">Networks</a>
21    <a href="#url">URL</a>
22    <a href="#domain">Domain</a>
23    <a href="#browser-server">Browser & Server</a>
24    <a href="#working">How a Website Works</a>
25    <a href="#client-server-script">Client vs Server Scripting</a>
26  </nav>
27
28  <main>
29
30  <!-- 1. Internet Overview -->
31  <section id="internet" class="topic">
32    <h2>1. Internet Overview</h2>
33    <p>The <strong>Internet</strong> is a global network of networks – millions of computers,
34    servers and devices connected together using a common set of rules (protocols) so they can
35    exchange data. It doesn't belong to one company or country; it's a cooperative system that
36    routes information from a source to a destination, usually in a fraction of a second.</p>
37    <p>The <strong>World Wide Web (WWW)</strong> is not the same thing as the Internet. The
38    Internet is the physical and logical network; the Web is one <em>service</em> that runs on
39    top of it – a collection of linked documents (webpages) accessed using HTTP/HTTPS. Email,
40    file transfer (FTP) and video calls are other services that also use the Internet.</p>
41
42    <div class="diagram">
43      <p class="diagram-label">Internet vs. Web</p>
44      <div class="diagram-row">
45        <div class="box box-wide">Internet <span class="box-sub">(the network – cables, routers,
46        ISPs)</span></div>
47      </div>
48      <div class="diagram-row three-up">
49        <div class="box">WWW <span class="box-sub">webpages</span></div>
50        <div class="box">Email <span class="box-sub">SMTP</span></div>
```

```

50         <div class="box">File Transfer <span class="box-sub">FTP</span></div>
51     </div>
52 </div>
53 </section>
54
55 <!-- 2. Fundamental Computer Network Concepts -->
56 <section id="network" class="topic">
57     <h2>2. Fundamental Computer Network Concepts</h2>
58     <p>A <strong>computer network</strong> is two or more devices connected so they can share
59     data and resources. A few core ideas every website team should know:</p>
60     <ul>
61         <li><strong>Node</strong> – any device on the network (a laptop, a server, a phone).</li>
62         <li><strong>IP Address</strong> – a numeric label (e.g. <code>192.168.1.10</code>) that
63         identifies a device on the network, similar to a postal address.</li>
64         <li><strong>Protocol</strong> – an agreed set of rules for communication, e.g.
65         <code>HTTP</code>, <code>TCP/IP</code>.</li>
66         <li><strong>LAN / WAN</strong> – a Local Area Network covers a small area (one campus
67         building); a Wide Area Network spans cities or countries (the Internet itself is the
68         largest WAN).</li>
69         <li><strong>Client-Server model</strong> – the client (browser) requests data; the
70         server stores and returns it. This is the model the entire Web is built on.</li>
71     </ul>
72
73     <div class="diagram">
74         <p class="diagram-label">Client-Server Communication</p>
75         <div class="diagram-row flow">
76             <div class="box">Client<span class="box-sub">Browser</span></div>
77             <div class="arrow">HTTP Request →</div>
78             <div class="box">Server<span class="box-sub">Web Server</span></div>
79         </div>
80         <div class="diagram-row flow">
81             <div class="box">Client<span class="box-sub">Browser</span></div>
82             <div class="arrow">← HTTP Response</div>
83             <div class="box">Server<span class="box-sub">Web Server</span></div>
84         </div>
85     </div>
86 </section>
87
88 <!-- 3. URL and its Structure -->
89 <section id="url" class="topic">
90     <h2>3. URL and its Structure</h2>
91     <p>A <strong>URL (Uniform Resource Locator)</strong> is the address typed into a browser to
92     reach a specific resource on the Web. It has several distinct parts:</p>
93
94     <div class="url-diagram">
95         <div class="url-string">
96             <span class="part scheme">https</span><span class="punct">://</span><span class="part
97             sub">www</span><span class="punct">.</span><span class="part domain">institution</span><span
98             class="punct">.</span><span class="part tld">edu</span><span class="part
99             path">/admissions/apply.html</span><span class="part query">?year=2026</span>
100         </div>
101         <div class="url-legend">
102             <div><span class="swatch scheme"></span> Scheme / Protocol – <code>https</code></div>
103             <div><span class="swatch sub"></span> Subdomain – <code>www</code></div>
104
105             <div><span class="swatch domain"></span> Domain name – <code>institution</code></div>

```

```

10     <div><span class="swatch tld"></span> Top-level domain – <code>.edu</code></div>
2
10     <div><span class="swatch path"></span> Path – <code>/admissions/apply.html</code></div>
3
10     <div><span class="swatch query"></span> Query string – <code>?year=2026</code></div>
4
10     </div>
5
10     </div>
6
10 </section>
7
10
8
10 <!-- 4. Domain Name -->
9
11 <section id="domain" class="topic">
0
11     <h2>4. Domain Name</h2>
1
11     <p>A <strong>domain name</strong> is the human-readable name that stands in for a server's
2
11     IP address (e.g. <code>institution.edu</code> instead of <code>203.0.113.42</code>). The
3
11     <strong>Domain Name System (DNS)</strong> is the internet's phone book – it translates the
4
11     domain name a person types into the numeric IP address a computer needs.</p>
5
11
6
11     <div class="diagram">
7
11         <p class="diagram-label">DNS Lookup</p>
8
11         <div class="diagram-row flow">
9
12             <div class="box">Browser<span class="box-sub">"institution.edu"</span></div>
0
12
12             <div class="arrow">→</div>
1
12
12             <div class="box">DNS Server<span class="box-sub">looks up the name</span></div>
2
12
12             <div class="arrow">→</div>
3
12
12             <div class="box">IP Address<span class="box-sub">203.0.113.42</span></div>
4
12
12         </div>
5
12
12     </div>
6
12 </section>
7
12
8
12 <!-- 5. Web Browsers and Web Servers -->
9
13 <section id="browser-server" class="topic">
0

```

```

<h2>5. Web Browsers and Web Servers</h2>

<div class="two-col">

  <div class="card">

    <h3>Web Browser</h3>

    <p>Software installed on the user's device (Chrome, Firefox, Edge) that sends
    requests for webpages and renders the HTML, CSS and JavaScript it receives into
    the page the visitor sees and interacts with.</p>

  </div>

  <div class="card">

    <h3>Web Server</h3>

    <p>Software (and often the machine running it) that stores website files and
    listens for incoming requests, sending back the correct page, image, or data –
    e.g. Apache, Nginx, or a Node.js server.</p>

  </div>

</div>

</section>

<!-- 6. Working Principle of a Website -->

<section id="working" class="topic">

  <h2>6. Working Principle of a Website</h2>

  <p>Putting the pieces above together, here is what happens when a visitor opens the
  institution's homepage:</p>

  <ol class="steps">

    <li>The visitor types <code>www.institution.edu</code> into the browser.</li>

    <li>The browser asks a DNS server to resolve that domain to an IP address.</li>

    <li>The browser sends an HTTP request to the web server at that IP address.</li>

    <li>The web server locates the requested file (e.g. <code>index.html</code>) and
    sends it back in an HTTP response.</li>

    <li>The browser parses the HTML/CSS/JavaScript and renders the finished page on
  
```

```

16         screen.</li>
0
16     </ol>
1
16 </section>
2
16
3
16 <!-- 7. Difference between Client-side and Server-side Scripting -->
4
16 <section id="client-server-script" class="topic">
5
16     <h2>7. Client-side vs. Server-side Scripting</h2>
6
16     <div class="table-wrap">
7
16         <table>
8
16             <caption>Where the code runs, and what it's good for</caption>
9
17             <thead>
0
17                 <tr>
1
17                     <th scope="col">Aspect</th>
2
17                     <th scope="col">Client-side Scripting</th>
3
17                     <th scope="col">Server-side Scripting</th>
4
17                 </tr>
5
17             </thead>
6
17             <tbody>
7
17                 <tr><td>Runs on</td><td>The visitor's browser</td><td>The web server</td></tr>
8
17                 <tr><td>Common languages</td><td>JavaScript</td><td>PHP, Node.js, Python, Java</td></tr>
9
18                 <tr><td>Typical use</td><td>Form validation, animations, interactivity</td><td>Database
0 access, authentication, generating pages dynamically</td></tr>
18                 <tr><td>Visibility</td><td>Source is visible to the user (View Source)</td><td>Source
1 stays private on the server</td></tr>
18                 <tr><td>Dependency</td><td>Needs the browser to execute</td><td>Runs regardless of the
2 visitor's device</td></tr>
18                 </tbody>
3
18             </table>
4
18         </div>
5
18     </section>
6
18
7
18 </main>
8

```

```

18
19
19 <footer class="page-footer">
20
19 <h2>Conclusion</h2>
21
19 <p>The Internet is the underlying network, and the Web is the linked-document service that
22
19 runs on it. A URL identifies a resource by combining a protocol, domain and path, and DNS
23
19 quietly translates that domain into an IP address for every request. Browsers and servers
24
19 talk to each other using the client-server model – the browser asks, the server answers –
25
19 and the page a visitor finally sees is the result of client-side code (JavaScript, for
26
19 interactivity) working alongside server-side code (for data and logic). Understanding this
27
19 chain is the first step before designing the institution's own site.</p>
28
19 </footer>
29
20 </body>
30
20 </html>
31

```

## ● style.css

```

1  /* =====
2   Task 1 – Understanding the World Wide Web
3   Simple, readable reference page with small CSS/box diagrams
4   ===== */
5
6  :root{
7    --ink:#1c2430;
8    --paper:#f6f5f1;
9    --panel:#ffffff;
10   --accent:#2f6fed;
11   --accent-soft:#e7effe;
12   --line:#d8d6cd;
13   --muted:#5c6470;
14 }
15
16 *{box-sizing:border-box;}
17 body{
18   margin:0;
19   font-family:'Segoe UI', system-ui, sans-serif;
20   background:var(--paper);
21   color:var(--ink);
22   line-height:1.6;
23 }
24
25 .page-header{
26   background:var(--ink);
27   color:#fff;

```

```

28     padding:40px 24px;
29     text-align:center;
30 }
31 .page-header h1{margin:0 0 8px;font-size:clamp(24px,4vw,36px);}
32 .subtitle{margin:0;color:#b9c2d0;font-size:15px;}
33
34 /* table-of-contents nav */
35 .toc{
36     display:flex;
37     flex-wrap:wrap;
38     gap:8px;
39     justify-content:center;
40     padding:16px;
41     background:var(--panel);
42     border-bottom:1px solid var(--line);
43     position:sticky;
44     top:0;
45     z-index:10;
46 }
47 .toc a{
48     color:var(--accent);
49     text-decoration:none;
50     font-size:13px;
51     padding:6px 12px;
52     border:1px solid var(--accent);
53     border-radius:20px;
54 }
55 .toc a:hover{background:var(--accent);color:#fff;}
56
57 main{max-width:900px;margin:0 auto;padding:20px 24px 40px;}
58
59 .topic{
60     background:var(--panel);
61     border:1px solid var(--line);
62     border-radius:10px;
63     padding:28px 30px;
64     margin:24px 0;
65 }
66 .topic h2{margin-top:0;color:var(--ink);border-left:4px solid var(--accent);padding-left:12px;}
67 .topic ul, .topic ol{padding-left:22px;}
68 .topic li{margin-bottom:6px;}
69 code{
70     background:var(--accent-soft);
71     color:#1a3f9c;
72     padding:2px 6px;
73     border-radius:4px;
74     font-size:0.92em;
75 }
76
77 /* generic box diagrams */
78 .diagram{margin-top:20px;}
79 .diagram-label{
80     font-size:12px;
81     text-transform:uppercase;
82     letter-spacing:0.08em;
83     color:var(--muted);
84     margin-bottom:10px;
85 }

```



```

86 .diagram-row{display:flex;gap:14px;flex-wrap:wrap;margin-bottom:10px;}
87 .diagram-row.three-up{justify-content:space-between;}
88 .diagram-row.flow{align-items:center;}
89 .box{
90   background:var(--accent-soft);
91   border:1.5px solid var(--accent);
92   border-radius:8px;
93   padding:14px 18px;
94   text-align:center;
95   font-weight:600;
96   flex:1;
97   min-width:140px;
98 }
99 .box-wide{flex:1 1 100%;}
100 .box-sub{display:block;font-weight:400;font-size:12px;color:var(--muted);margin-top:4px;}
101
102 .arrow{font-family:'Consolas', monospace;color:var(--muted);white-space:nowrap;}
103
104
105
106 /* URL structure diagram */
107
108 .url-diagram{margin-top:18px;}
109
110 .url-string{
111
112   font-family:'Consolas', monospace;
113
114   font-size:clamp(13px,2.4vw,17px);
115
116   background:var(--ink);
117
118   color:#e8ecf3;
119
120   padding:16px 18px;
121
122   border-radius:8px;
123
124   overflow-x:auto;
125
126   white-space:nowrap;
127
128   margin-bottom:16px;
129 }
130
131 .url-string .part{padding:2px 3px;border-radius:3px;}
132
133 .punct{color:#8b93a3;}
134
135 .scheme{background:#2f6fed55;color:#8fb4ff;}
136
137 .sub{background:#37b06455;color:#8fe0a8;}
138
139 .domain{background:#e1a53a55;color:#ffd486;}
140
141 .tld{background:#e1a53a55;color:#ffd486;}

```

```

12 .path{background:#c05aed55;color:#e0aaff;}
2
12 .query{background:#e0505055;color:#ff9d9d;}
3
12
4
12 .url-legend{
5
12     display:grid;
6
12     grid-template-columns:repeat(auto-fit,minmax(220px,1fr));
7
12     gap:8px 20px;
8
12     font-size:14px;
9
13 }
0
13 .url-legend div{display:flex;align-items:center;gap:8px;}
1
13 .swatch{width:14px;height:14px;border-radius:3px;display:inline-block;flex-shrink:0;}
2
13 .swatch.scheme{background:#8fb4ff;}
3
13 .swatch.sub{background:#8fe0a8;}
4
13 .swatch.domain,.swatch.tld{background:#ffd486;}
5
13 .swatch.path{background:#e0aaff;}
6
13 .swatch.query{background:#ff9d9d;}
7
13
8
13 /* two-column cards (browser vs server) */
9
14 .two-col{display:grid;grid-template-columns:1fr 1fr;gap:20px;margin-top:16px;}
0
14 .card{background:var(--accent-soft);border-radius:8px;padding:18px 20px;}
1
14 .card h3{margin-top:0;color:#1a3f9c;}
2
14
3
14 /* numbered steps */
4
14 .steps li{margin-bottom:10px;}
5
14
6
14 /* table */
7
14 .table-wrap{overflow-x:auto;margin-top:16px;}
8
14 table{border-collapse:collapse;width:100%;min-width:560px;}
9
15 caption{caption-side:top;text-align:left;font-size:13px;color:var(--muted);padding-bottom:8px;}
0

```

```
15 th,td{text-align:left;padding:10px 14px;border-bottom:1px solid var(--line);font-size:14px;}
1
15 thead th{background:var(--ink);color:#fff;}
2
15 tbody tr:nth-child(even){background:#fbfaf7;}
3
15
4
15 /* footer / conclusion */
5
15 .page-footer{
6
15     max-width:900px;
7
15     margin:0 auto 50px;
8
15     padding:24px 30px;
9
16     background:var(--accent-soft);
0
16     border-radius:10px;
1
16 }
2
16 .page-footer h2{margin-top:0;color:#1a3f9c;}
3
16
4
16 @media (max-width:700px){
5
16     .two-col{grid-template-columns:1fr;}
6
16     .diagram-row.three-up{flex-direction:column;}
7
16 }
8
```

# Task 2 — Student Admission Form

Folder: Assignment-WebTech-2026-08-18/Task2-Admission-Form/

● index.html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Student Admission Form</title>
7  <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10
11  <header class="site-header">
12    <h1>Online Admission Form – 2026 Batch</h1>
13    <p>Fill in your details below. Fields marked <span class="req">*</span> are required.</p>
14  </header>
15
16  <main class="page-wrap">
17
18    <!-- Short intro video: a media element (video) demonstrating the campus -->
19    <section class="intro-media">
20      <h2>Take a quick campus tour before you apply</h2>
21      <!-- controls attribute gives the visitor play/pause/volume controls -->
22      <video controls
poster="
iIGhlaWdodD0iNDAwIj4KPHJlY3Qgd2lkdGg9IjgwMCIgaGVPZ2h0PSI0MDAiIGZpbGw9IiMxYzI0MzAiLz4KPHJlY3QgeD0iMCIg
eT0iMCIgd2lkdGg9IjgwMCIgaGVPZ2h0PSI0MDAiIGZpbGw9InVybCg9IjZyKiLz4KPGRlZnM+PGxpbmVhckdyYWVpZ2h0PSI0MDA
iB4MT0iMCIgeTE9IjAiIHgyPSI0IiB5Mj0iMSI+CjxzZDg9wIG9mZnNldD0iMCIgc3RvcC1jb2xvcj0iIzJmNmZlZCIgc3RvcC1vcG
FjaXR5PSIwLjM1Ii8+CjxzZDg9wIG9mZnNldD0iMSIgc3RvcC1jb2xvcj0iIzFjMjQzMCIgc3RvcC1vcGFjaXR5PSIwLjM1Ii8+Cjwv
uZWYr3JhZG1lbnQ+PC9kZWZzPgo8Y2lyY2x1IGN4PSI0MDAiIGN5PSI0TAiIH9IjQ2IiBmaWxsPSIjZmZmZmZmIiBmaWxsLW9w
YWNpdHk9IjAuODUiLz4KPHBvbHlnb24gcG9pbmRzPSIzODgsMTY4IDM4OCwyMTIgNDI0LDE5MCIgZmlsbD0iIzFjMjQzMCIvPgo8d
GV4dCB4PSI0MDAiIHk9IjI3MCIgZm9udC1mYW1pbHk9IiNlZ29lIFVJLCBzYW5zLXNlcm1mIiBmb250LXNpemU9IjE4IiBmaWxsPS
IjZmZmZmZmIiB0ZXh0LWFuY2hvcj0ibWlkZGx1Ij5DYW1wdXMgVG91ciDigJQgdmlkZW8gcGxhY2Vob2xkZXI8L3RleHQ+Cjwvc3Z
nPg==" width="100%">
23      Your browser does not support the video tag. A real campus-tour.mp4 file would be linked
here.
24    </video>
25
26    <!-- Audio media element: a short welcome message from the admissions office -->
27    <p class="audio-label">Listen: a short welcome message from the Admissions Office</p>
28    <audio controls>
29      <source src="welcome-message.mp3" type="audio/mpeg">
30      Your browser does not support the audio tag.
31    </audio>
32  </section>
33
34  <form class="admission-form" id="admissionForm">
35
36    <!-- ===== Personal Details ===== -->
37    <fieldset>
38      <legend>Personal Details</legend>
39
```

```
<div class="field">
    <label for="fullName">Full Name <span class="req">*</span></label>
    <!-- text input -->
    <input type="text" id="fullName" name="fullName" placeholder="As per school records"
required>
</div>

<div class="field-row">
    <div class="field">
        <label for="dob">Date of Birth <span class="req">*</span></label>
        <!-- date input type -->
        <input type="date" id="dob" name="dob" required>
    </div>
    <div class="field">
        <label for="age">Age</label>
        <!-- number input type, restricted with min/max -->
        <input type="number" id="age" name="age" min="14" max="25" placeholder="Years">
    </div>
</div>

<div class="field">
    <label>Gender <span class="req">*</span></label>
    <!-- radio button group -->
    <div class="radio-row">
        <label class="inline"><input type="radio" name="gender" value="female" required>
Female</label>
        <label class="inline"><input type="radio" name="gender" value="male"> Male</label>
        <label class="inline"><input type="radio" name="gender" value="other"> Other</label>
    </div>
</div>

<div class="field">
    <label for="photo">Passport Size Photo</label>
    <!-- file input, restricted to images -->
    <input type="file" id="photo" name="photo" accept="image/*">
</div>

<!-- Sample profile image already on the page – a media element (image) -->
<div class="field">
    <label>Sample Format</label>
    
    </div>
</fieldset>

<!-- ===== Contact Details ===== -->
<fieldset>
    <legend>Contact Details</legend>

    <div class="field-row">
        <div class="field">
            <label for="email">Email Address <span class="req">*</span></label>
            <!-- email input, browser validates the format -->
```

```

90         <input type="email" id="email" name="email" placeholder="you@example.com" required>
91     </div>
92     <div class="field">
93         <label for="phone">Phone Number <span class="req">*</span></label>
94         <!-- tel input -->
95         <input type="tel" id="phone" name="phone" pattern="[0-9]{10}" placeholder="10-digit
mobile number" required>
96     </div>
97 </div>
98
99     <div class="field">
100         <label for="address">Postal Address</label>
101
102         <textarea id="address" name="address" rows="3" placeholder="House no, street, city, state,
PIN"></textarea>
103     </div>
104
105     <div class="field">
106
107         <label for="website">Portfolio / LinkedIn (optional)</label>
108
109         <!-- url input -->
110
111         <input type="url" id="website" name="website" placeholder="https://">
112
113     </div>
114 </fieldset>
115
116 <!-- ===== Course Details ===== -->
117
118 <fieldset>
119
120     <legend>Course Details</legend>
121
122     <div class="field">
123
124         <label for="course">Course Applying For <span class="req">*</span></label>
125
126         <!-- select dropdown -->
127
128         <select id="course" name="course" required>
129
130             <option value="" disabled selected>Choose a course</option>
131
132             <option value="cse">B.Tech – Computer Science</option>
133
134             <option value="ece">B.Tech – Electronics & Communication</option>
135
136             <option value="mech">B.Tech – Mechanical</option>

```

```

12         <option value="mba">MBA</option>
3
12     </select>
4
12 </div>
5
12
6
12     <div class="field">
7
12         <label>Preferred Shift</label>
8
12         <div class="radio-row">
9
13             <label class="inline"><input type="radio" name="shift" value="morning" checked>
0 Morning</label>
13             <label class="inline"><input type="radio" name="shift" value="evening"> Evening</label>
1
13         </div>
2
13     </div>
3
13
4
13     <div class="field">
5
13         <label>Hostel Required?</label>
6
13         <!-- checkbox: single boolean choice -->
7
13         <label class="inline"><input type="checkbox" id="hostel" name="hostel"> Yes, I need hostel
8 accommodation</label>
13     </div>
9
14
0
14     <div class="field">
1
14         <label for="marks">Percentage in Qualifying Exam <span class="req">*</span></label>
2
14         <!-- range input paired with a live number readout -->
3
14         <input type="range" id="marks" name="marks" min="0" max="100" value="75"
4
14             oninput="marksValue.value = marks.value + '%">
5
14         <output id="marksValue" for="marks">75%</output>
6
14     </div>
7
14 </fieldset>
8
14
9
15 <!-- ===== Declaration ===== -->
0
15 <fieldset>
1

```

```

15     <legend>Declaration</legend>
2
15     <label class="inline">
3
15         <input type="checkbox" id="declare" name="declare" required>
4
15         I confirm the details provided above are true to the best of my knowledge. <span
5 class="req">*</span>
15     </label>
6
15 </fieldset>
7
15
8
15 <div class="form-actions">
9
16     <button type="submit" class="btn-primary">Submit Application</button>
0
16     <button type="reset" class="btn-ghost">Clear Form</button>
1
16 </div>
2
16
3
16     <p id="successMsg" class="success-msg">Application submitted – thank you! (Demo only, no data
4 is sent anywhere.)</p>
16 </form>
5
16 </main>
6
16
7
16 <footer class="page-footer">
8
16     <p>© 2026 Admissions Office. Demo form for coursework – not a live application system.</p>
9
17 </footer>
0
17
1
17 <script>
2
17     // Prevent an actual page reload on submit – this is a static demo form
3
17     document.getElementById('admissionForm').addEventListener('submit', function (e) {
4
17         e.preventDefault();
5
17         document.getElementById('successMsg').classList.add('show');
6
17         this.reset();
7
17         marksValue.value = '75%'; // reset the range readout back to its default
8
17     });
9
18 </script>
0

```



```
18
1
18 </body>
2
18 </html>
3
```

## ● style.css

```
1  /* =====
2    Task 2 – Responsive Student Admission Form
3    Demonstrates: CSS3 selectors, the box model, and a responsive
4    layout that reflows from a 2-column grid to a single column.
5    ===== */
6
7  /* ---- CSS3 Selector: universal selector, resets the box model ---- */
8  *{
9    box-sizing:border-box; /* padding/border counted INSIDE width – core box-model choice */
10   margin:0;
11 }
12
13 :root{
14   --ink:#1c2430;
15   --paper:#f4f6fb;
16   --panel:#ffffff;
17   --accent:#2f6fed;
18   --accent-dark:#1a4fc4;
19   --line:#d7dce6;
20   --danger:#e15050;
21 }
22
23 /* ---- Type selector ---- */
24 body{
25   font-family:'Segoe UI', system-ui, sans-serif;
26   background:var(--paper);
27   color:var(--ink);
28   line-height:1.5;
29 }
30
31 .site-header{
32   background:var(--ink);
33   color:#fff;
34   padding:32px 20px;
35   text-align:center;
36 }
37 .site-header h1{font-size:clamp(20px,4vw,30px);margin-bottom:6px;}
38
39 /* ---- Attribute selector: styles only <span> elements with class "req" ---- */
40 span.req{color:#ff8080;font-weight:700;}
41
42 .page-wrap{
43   max-width:760px;
44   margin:0 auto;
45   padding:24px 16px 60px;
46 }
47
48 /* ---- Descendant selector ---- */
49 .intro-media h2{font-size:17px;margin-bottom:10px;}
```

```

50 .intro-media video, .intro-media audio{
51     width:100%;
52     border-radius:10px;
53     outline:1px solid var(--line);
54 }
55 .intro-media audio{width:100%;margin-top:4px;}
56 .audio-label{font-size:14px;color:#4a5468;margin:18px 0 6px;}
57 .intro-media{margin-bottom:30px;}
58
59 /* ---- BOX MODEL DEMONSTRATION ----
60     Each .field wraps a label+input pair. The visible box below is
61     built from: content (input) -> padding -> border -> margin.
62 */
63 form.admission-form{
64     background:var(--panel);
65     border-radius:12px;
66     border:1px solid var(--line);
67     padding:8px;
68 }
69
70 fieldset{
71     border:1px solid var(--line); /* border edge of the box */
72     border-radius:10px;
73     padding:22px 24px; /* padding edge of the box */
74     margin:18px; /* margin edge of the box */
75 }
76 legend{
77     padding:0 10px; /* padding around legend text */
78     font-weight:700;
79     color:var(--accent-dark);
80     font-size:15px;
81 }
82
83 .field{margin-bottom:18px;} /* margin between stacked fields */
84 .field:last-child{margin-bottom:0;}
85 .field label{
86     display:block;
87     font-size:13px;
88     font-weight:600;
89     margin-bottom:6px;
90     color:#3a4358;
91 }
92
93 /* text-like inputs share one box-model treatment via a grouped selector */
94 input[type="text"],
95 input[type="date"],
96 input[type="number"],
97 input[type="email"],
98 input[type="tel"],
99 input[type="url"],
100 select,
101
102
103 textarea{
104
105     width:100%;
106
107     padding:11px 13px; /* content sits inside this padding */
108
109 }

```

```

10 border:1.5px solid var(--line); /* border wraps the padding */
4
10 border-radius:7px;
5
10 font-size:14px;
6
10 font-family:inherit;
7
10 background:#fbfcfe;
8
10 transition:border-color .15s ease, box-shadow .15s ease;
9
11 }
0
11
1
11 /* ---- Pseudo-class selector: focus state ---- */
2
11 input:focus, select:focus, textarea:focus{
3
11     outline:none;
4
11     border-color:var(--accent);
5
11     box-shadow:0 0 3px rgba(47,111,237,0.15);
6
11 }
7
11
8
11 /* ---- Pseudo-class selector: validity state ---- */
9
12 input:required:invalid{border-color:var(--line);}
0
12 input:not(:placeholder-shown):invalid{border-color:var(--danger);}
1
12
2
12 .field-row{
3
12     display:grid;
4
12     grid-template-columns:1fr 1fr;
5
12     gap:16px;
6
12 }
7
12
8
12 .radio-row{display:flex;gap:18px;flex-wrap:wrap;}
9
13 label.inline{
0
13     display:flex;
1
13     align-items:center;
2

```

```
13     gap:8px;
3
13     font-weight:400;
4
13     font-size:14px;
5
13     color:#3a4358;
6
13     cursor:pointer;
7
13 }
8
13 label.inline input{width:auto;}
9
14
0
14 input[type="range"]{width:80%;vertical-align:middle;}
1
14 output{font-weight:700;color:var(--accent-dark);margin-left:10px;}
2
14
3
14 .sample-photo{
4
14     width:100px;
5
14     border:2px solid var(--line);
6
14     border-radius:6px;
7
14     padding:4px;           /* padding around the image inside its border */
8
14 }
9
15
0
15 .form-actions{
1
15     display:flex;
2
15     gap:14px;
3
15     padding:20px 24px 24px;
4
15     flex-wrap:wrap;
5
15 }
6
15 .btn-primary, .btn-ghost{
7
15     padding:13px 26px;
8
15     border-radius:8px;
9
15     font-size:14px;
0
16     font-weight:700;
1
16 }
```

```

16     cursor:pointer;
2
16     border:2px solid var(--accent);
3
16 }
4
16 .btn-primary{background:var(--accent);color:#fff;}
5
16 .btn-primary:hover{background:var(--accent-dark);border-color:var(--accent-dark);}
6
16 .btn-ghost{background:transparent;color:var(--accent-dark);}
7
16 .btn-ghost:hover{background:var(--accent);color:#fff;}
8
16
9
17 .success-msg{
0
17     display:none;
1
17     margin:0 24px 22px;
2
17     padding:14px 18px;
3
17     background:#e5f6ea;
4
17     border:1px solid #7fd39a;
5
17     color:#1a6b34;
6
17     border-radius:8px;
7
17     font-size:14px;
8
17 }
9
18 .success-msg.show{display:block;}
0
18
1
18 .page-footer{
2
18     text-align:center;
3
18     padding:20px;
4
18     font-size:13px;
5
18     color:#6b7386;
6
18 }
7
18
8
18 /* ----- RESPONSIVE BREAKPOINTS ----- */
9
19 @media (max-width:640px){
0

```

```
19 .field-row{grid-template-columns:1fr;}
1
19 fieldset{margin:12px;padding:18px 16px;}
2
19 .form-actions{flex-direction:column;}
3
19 .btn-primary, .btn-ghost{width:100%;}
4
19 }
5
```

# Task 3 — Online Fashion Store Homepage

Folder: Assignment-WebTech-2026-08-18/Task3-Fashion-Homepage/

● index.html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>LUXE & LINEN – Online Fashion Store</title>
7  <link rel="preconnect" href="https://fonts.googleapis.com">
8  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
9  <link href="https://fonts.googleapis.com/css2?family=Fraunces:opsz,wght@9..144,400;9..144,600;9..144,700&family=Work+Sans:wght@400;500;600;700&display=swap" rel="stylesheet">
10 <link rel="stylesheet" href="style.css">
11 </head>
12 <body>
13
14   <header class="site-header">
15     <div class="nav-wrap">
16       <span class="logo">LUXE <span class="amp">&</span> LINEN</span>
17       <nav>
18         <a href="#new">New In</a>
19         <a href="#bestsellers">Bestsellers</a>
20         <a href="#about">About</a>
21       </nav>
22     </div>
23   </header>
24
25   <!-- ===== HERO: backgrounds, borders, text effects, animation ===== -->
26   <section class="hero">
27     <div class="hero-copy">
28       <p class="eyebrow">Autumn / Winter 2026</p>
29       <h1>Wear the<br><span class="highlight">season</span> well.</h1>
30       <p class="hero-desc">Considered pieces, honest fabrics, and a fit that holds up past the first wash. New arrivals drop every Friday.</p>
31       <a href="#bestsellers" class="btn">Shop the edit</a>
32     </div>
33     <div class="hero-visual" aria-hidden="true">
34       <div class="floating-card card-1">100% Linen</div>
35       <div class="floating-card card-2">Free returns</div>
36       <div class="floating-card card-3">★ 4.8 rated</div>
37     </div>
38   </section>
39
40   <!-- ===== MULTI-COLUMN LAYOUT: brand values ===== -->
41   <section class="values" id="about">
42     <h2>Why shop with us</h2>
43     <div class="value-columns">
44       <p>Every fabric is sourced from mills we've visited ourselves – no blind third-party sourcing, ever. We believe the label should tell the truth about where a garment came from.</p>
45       <p>Sizing runs true across the whole catalog, tested on real bodies, not a single sample fit model. If something runs differently, the product page says so plainly.</p>
```

```

46     <p>Free exchanges within 30 days, no receipt required if you checked out with an account.
Returns are processed within 48 hours of the item reaching our warehouse.</p>
47     <p>A small in-house tailoring team can adjust hems and waists on request for a flat fee – a
service most online stores don't offer at all.</p>
48     </div>
49 </section>
50
51 <!-- ===== PRODUCT GRID: cards with border/background/text treatments ===== -->
52 <section class="collection" id="new">
53     <h2>New In</h2>
54     <div class="product-grid">
55
56         <article class="product-card">
57             <div class="product-media tone-clay">
58                 <span class="tag">New</span>
59             </div>
60             <h3>Linen Wrap Dress</h3>
61             <p class="price">₹3,450</p>
62         </article>
63
64         <article class="product-card">
65             <div class="product-media tone-forest">
66                 <span class="tag">Limited</span>
67             </div>
68             <h3>Wool Overcoat</h3>
69             <p class="price">₹7,200</p>
70         </article>
71
72         <article class="product-card">
73             <div class="product-media tone-sand">
74             </div>
75             <h3>Cotton Shirt – Relaxed</h3>
76             <p class="price">₹2,100</p>
77         </article>
78
79         <article class="product-card">
80             <div class="product-media tone-rust">
81                 <span class="tag">New</span>
82             </div>
83             <h3>Corduroy Trousers</h3>
84             <p class="price">₹2,850</p>
85         </article>
86
87     </div>
88 </section>
89
90 <!-- ===== BESTSELLERS: animated marquee strip ===== -->
91 <section class="marquee-wrap" id="bestsellers">
92     <div class="marquee">
93         <div class="marquee-track">
94             <span>Linen Wrap Dress</span><span>.</span>
95             <span>Wool Overcoat</span><span>.</span>
96             <span>Cotton Shirt</span><span>.</span>
97             <span>Corduroy Trousers</span><span>.</span>
98             <span>Silk Scarf</span><span>.</span>
99             <span>Linen Wrap Dress</span><span>.</span>
100            <span>Wool Overcoat</span><span>.</span>

```



```

10     <span>Cotton Shirt</span><span>·</span>
1
10     <span>Corduroy Trousers</span><span>·</span>
2
10     <span>Silk Scarf</span><span>·</span>
3
10     </div>
4
10     </div>
5
10 </section>
6
10
7
10 <footer class="site-footer">
8
10     <p>© 2026 Luxe & Linen – student coursework demo, not a real store.</p>
9
11 </footer>
0
11
1
11 </body>
2
11 </html>
3

```

## ● style.css

```

1  /* =====
2   Task 3 – Online Fashion Store Homepage
3   Demonstrates: backgrounds & borders, text effects, CSS
4   animations, a multi-column layout, and basic UI principles
5   (hierarchy, consistency, affordance, whitespace).
6   ===== */
7
8  :root{
9      --ink:#1b2320;          /* near-black with a green undertone, not pure black */
10     --paper:#eef0e8;        /* pale sage-grey background, a considered neutral */
11     --panel:#ffffff;
12     --accent:#2f6b4a;       /* forest green – signature accent */
13     --accent-soft:#dfe8de;
14     --gold:#c98a3b;         /* warm ochre – secondary accent for price/tags */
15     --line:rgba(27,35,32,0.16);
16     --muted:#5b6660;
17     --display:'Fraunces', Georgia, serif;
18     --body:'Work Sans', system-ui, sans-serif;
19 }
20
21 *{box-sizing:border-box;}
22 html{scroll-behavior:smooth;}
23 body{
24     margin:0;
25     background:var(--paper);
26     color:var(--ink);
27     font-family:var(--body);
28     line-height:1.6;
29 }

```

```

30 h1,h2,h3{font-family:var(--display);margin:0;line-height:1.1;}
31 a{color:inherit;}
32
33 /* ----- header ----- */
34 .site-header{
35     position:sticky;top:0;z-index:20;
36     background:rgba(238,240,232,0.92);
37     backdrop-filter:blur(6px);
38     border-bottom:1px solid var(--line);
39 }
40 .nav-wrap{
41     max-width:1180px;margin:0 auto;
42     display:flex;align-items:center;justify-content:space-between;
43     padding:20px 28px;
44 }
45 .logo{font-family:var(--display);font-size:20px;letter-spacing:0.03em;font-weight:600;}
46 .logo .amp{color:var(--gold);font-style:italic;}
47 .site-header nav{display:flex;gap:30px;}
48 .site-header nav a{
49     text-decoration:none;font-size:14px;font-weight:500;
50     position:relative;padding-bottom:4px;
51 }
52 /* text effect: animated underline reveal on hover (UI affordance) */
53 .site-header nav a::after{
54     content:"";position:absolute;left:0;bottom:0;width:100%;height:1.5px;
55     background:var(--accent);transform:scaleX(0);transform-origin:right;
56     transition:transform .25s ease;
57 }
58 .site-header nav a:hover::after{transform:scaleX(1);transform-origin:left;}
59
60 /* ----- hero: background + text effects + animation ----- */
61 .hero{
62     position:relative;
63     max-width:1180px;margin:0 auto;
64     display:grid;grid-template-columns:1.1fr 0.9fr;gap:40px;
65     align-items:center;
66     padding:80px 28px 90px;
67     overflow:hidden;
68 }
69 .eyebrow{
70     font-size:13px;letter-spacing:0.14em;text-transform:uppercase;
71     color:var(--accent);font-weight:600;margin-bottom:16px;
72 }
73 .hero h1{font-size:clamp(42px,6vw,72px);font-weight:600;}
74 /* text effect: gradient-clipped highlight word */
75 .highlight{
76     background:linear-gradient(100deg, var(--accent) 0%, var(--gold) 100%);
77     -webkit-background-clip:text;
78     background-clip:text;
79     color:transparent;
80     font-style:italic;
81 }
82 .hero-desc{
83     max-width:42ch;font-size:17px;color:#3d4740;
84     margin:22px 0 30px;
85 }
86 .btn{
87     display:inline-block;

```

```

88     background:var(--ink);color:var(--paper);
89     padding:15px 30px;
90     text-decoration:none;
91     font-size:14px;font-weight:600;letter-spacing:0.03em;
92     border:1.5px solid var(--ink);
93     border-radius:2px;
94     transition:transform .18s ease, background .18s ease, color .18s ease;
95 }
96 .btn:hover{background:transparent;color:var(--ink);transform:translateY(-2px);}
97
98 /* hero visual: background pattern + border + floating-card animation */
99 .hero-visual{
100     position:relative;height:380px;
101
102     background:
103         radial-gradient(circle at 30% 20%, var(--accent-soft) 0%, transparent 55%),
104         radial-gradient(circle at 75% 70%, #f3e6ce 0%, transparent 50%),
105         var(--panel);
106
107     border:1.5px solid var(--ink);
108
109     border-radius:4px;
110 }
111
112 .floating-card{
113     position:absolute;
114
115     background:var(--panel);
116
117     border:1px solid var(--ink);
118
119     padding:10px 16px;
120
121     font-size:13px;font-weight:600;
122
123     border-radius:3px;
124
125     box-shadow:4px 4px 0 rgba(27,35,32,0.12);
126
127     animation:float 5s ease-in-out infinite;
128 }
129
130 .card-1{top:14%;left:10%;animation-delay:0s;}
131
132 .card-2{top:52%;right:8%;animation-delay:1.2s;color:var(--accent);}
133
134 .card-3{bottom:10%;left:22%;animation-delay:2.4s;color:var(--gold);}
135
136 /* CSS animation: gentle vertical float, respects reduced motion below */
137 @keyframes float{
138

```

```

12 0%,100%{transform:translateY(0);}
3
12 50%{transform:translateY(-12px);}
4
12 }
5
12
6
12 /* ----- values: MULTI-COLUMN LAYOUT ----- */
7
12 .values{max-width:1180px;margin:0 auto;padding:70px 28px;}
8
12 .values h2{font-size:clamp(26px,3.4vw,36px);margin-bottom:28px;}
9
13 .value-columns{
0
13 column-count:2;
1 column-gap:48px;
13 column-rule:1px solid var(--line);
2
13 font-size:15px;
4
13 color:#333d36;
5
13 }
6
13 .value-columns p{margin:0 0 20px;break-inside:avoid;}
7
13
8
13 /* ----- product grid: backgrounds & borders ----- */
9
14 .collection{max-width:1180px;margin:0 auto;padding:40px 28px 80px;}
0
14 .collection h2{font-size:clamp(26px,3.4vw,36px);margin-bottom:30px;}
1
14 .product-grid{
2
14 display:grid;grid-template-columns:repeat(auto-fit,minmax(230px,1fr));
3
14 gap:26px;
4
14 }
5
14 .product-card{
6
14 background:var(--panel);
7
14 border:1px solid var(--line);
8
14 border-radius:4px;
9
15 overflow:hidden;
0
15 transition:transform .2s ease, box-shadow .2s ease;
1

```

```

15 }
2
15 .product-card:hover{transform:translateY(-5px);box-shadow:0 14px 26px -14px rgba(27,35,32,0.35);}
3
15 .product-media{
4
15     position:relative;
5
15     height:220px;
6
15     border-bottom:1px solid var(--line);
7
15 }
8
15 /* backgrounds: distinct gradient "fabric tones" per product, no photos needed */
9
16 .tone-clay{background:linear-gradient(160deg,#d9a679,#b97a55);}
0
16 .tone-forest{background:linear-gradient(160deg,#4b7a5c,#274a37);}
1
16 .tone-sand{background:linear-gradient(160deg,#e7dcc4,#c9b98f);}
2
16 .tone-rust{background:linear-gradient(160deg,#a8532f,#7a3820);}
3
16 .product-card .tag{
4
16     position:absolute;top:12px;left:12px;
5
16     background:var(--ink);color:var(--paper);
6
16     font-size:11px;font-weight:600;letter-spacing:0.05em;text-transform:uppercase;
7
16     padding:5px 10px;border-radius:2px;
8
16 }
9
17 .product-card h3{font-size:16px;font-weight:600;padding:16px 16px 4px;}
0
17 .product-card .price{
1
17     font-family:var(--body);font-weight:700;color:var(--gold);
2
17     padding:0 16px 18px;font-size:15px;
3
17 }
4
17
5
17 /* ----- marquee: continuous CSS animation ----- */
6
17 .marquee-wrap{
7
17     background:var(--ink);color:var(--paper);
8
17     padding:22px 0;overflow:hidden;
9
18     border-top:1px solid var(--ink);border-bottom:1px solid var(--ink);
0

```

```

18 }
1
18 .marquee-track{
2
18     display:inline-flex;gap:14px;white-space:nowrap;
3
18     font-family:var(--display);font-size:20px;font-style:italic;
4
18     animation:scroll-left 22s linear infinite;
5
18     padding-left:100%;
6
18 }
7
18 .marquee-track span{opacity:0.9;}
8
18 @keyframes scroll-left{
9
19     from{transform:translateX(0);}
0
19     to{transform:translateX(-100%);}
1
19 }
2
19 @media (prefers-reduced-motion: reduce){
3
19     .floating-card, .marquee-track{animation:none;}
4
19 }
5
19
6
19 .site-footer{
7
19     text-align:center;padding:26px;font-size:13px;color:var(--muted);
8
19 }
9
20
0
20 /* ----- responsive ----- */
1
20 @media (max-width:860px){
2
20     .hero{grid-template-columns:1fr;padding-top:50px;}
3
20     .hero-visual{height:260px;}
4
20     .value-columns{column-count:1;}
5
20 }
6

```

# Task 4 — Student Grade Calculator

Folder: Assignment-WebTech-2026-08-18/Task4-Grade-Calculator/

● index.html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Student Grade Calculator</title>
7  <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10
11  <main class="calc-wrap">
12    <h1>Grade Calculator</h1>
13    <p class="subtitle">Enter marks (out of 100) for five subjects to get the total, average, grade
    and pass/fail status.</p>
14
15    <form id="gradeForm">
16      <div class="subject-grid">
17        <div class="field">
18          <label for="s1">Subject 1</label>
19          <input type="number" id="s1" min="0" max="100" value="0" required>
20        </div>
21        <div class="field">
22          <label for="s2">Subject 2</label>
23          <input type="number" id="s2" min="0" max="100" value="0" required>
24        </div>
25        <div class="field">
26          <label for="s3">Subject 3</label>
27          <input type="number" id="s3" min="0" max="100" value="0" required>
28        </div>
29        <div class="field">
30          <label for="s4">Subject 4</label>
31          <input type="number" id="s4" min="0" max="100" value="0" required>
32        </div>
33        <div class="field">
34          <label for="s5">Subject 5</label>
35          <input type="number" id="s5" min="0" max="100" value="0" required>
36        </div>
37      </div>
38
39      <button type="submit">Calculate Result</button>
40    </form>
41
42    <section id="result" class="result-panel hidden">
43      <div class="result-row">
44        <span>Total</span>
45        <strong id="rTotal">0</strong>
46      </div>
47      <div class="result-row">
48        <span>Average</span>
49        <strong id="rAverage">0%</strong>
```

```

50     </div>
51     <div class="result-row">
52         <span>Grade</span>
53         <strong id="rGrade">-</strong>
54     </div>
55     <div class="result-row">
56         <span>Status</span>
57         <strong id="rStatus" class="status">-</strong>
58     </div>
59 </section>
60
61     <p id="errorMsg" class="error-msg hidden">Please enter a valid mark between 0 and 100 for every
subject.</p>
62 </main>
63
64     <script src="script.js"></script>
65 </body>
66 </html>

```

## ● style.css

```

1  /* Task 4 – Grade Calculator styling */
2
3  :root{
4      --ink:#20242b;
5      --paper:#f2f4f8;
6      --panel:#ffffff;
7      --accent:#3556d1;
8      --line:#dbe0ea;
9      --pass:#1f8a4c;
10     --fail:#c8402f;
11 }
12
13 *{box-sizing:border-box;}
14 body{
15     margin:0;
16     min-height:100vh;
17     display:flex;
18     align-items:center;
19     justify-content:center;
20     background:var(--paper);
21     font-family:'Segoe UI', system-ui, sans-serif;
22     color:var(--ink);
23     padding:30px 16px;
24 }
25
26 .calc-wrap{
27     width:100%;
28     max-width:440px;
29     background:var(--panel);
30     border:1px solid var(--line);
31     border-radius:12px;
32     padding:32px 30px;
33     box-shadow:0 20px 40px -28px rgba(20,25,40,0.35);
34 }
35
36 h1{margin:0 0 6px;font-size:24px;}
37 .subtitle{margin:0 0 24px;font-size:14px;color:#5a6072;}

```



```

38
39 .subject-grid{
40     display:grid;
41     grid-template-columns:1fr 1fr;
42     gap:14px;
43     margin-bottom:22px;
44 }
45 .field label{
46     display:block;font-size:12px;font-weight:600;
47     text-transform:uppercase;letter-spacing:0.04em;
48     color:#5a6072;margin-bottom:6px;
49 }
50 .field input{
51     width:100%;padding:10px 12px;
52     border:1.5px solid var(--line);border-radius:7px;
53     font-size:15px;font-family:inherit;
54 }
55 .field input:focus{outline:none;border-color:var(--accent);}
56
57 button[type="submit"]{
58     width:100%;padding:13px;
59     background:var(--accent);color:#fff;
60     border:none;border-radius:8px;
61     font-size:15px;font-weight:700;cursor:pointer;
62 }
63 button[type="submit"]:hover{background:#2a44ac;}
64
65 .result-panel{
66     margin-top:24px;padding:18px 20px;
67     background:var(--paper);border-radius:10px;
68     border:1px solid var(--line);
69 }
70 .result-row{
71     display:flex;justify-content:space-between;
72     padding:8px 0;font-size:14px;
73     border-bottom:1px dashed var(--line);
74 }
75 .result-row:last-child{border-bottom:none;}
76 .status.pass{color:var(--pass);}
77 .status.fail{color:var(--fail);}
78
79 .error-msg{
80     margin-top:18px;padding:12px 14px;
81     background:#fbe9e7;color:var(--fail);
82     border:1px solid #f0b6ab;border-radius:8px;
83     font-size:13px;
84 }
85 .hidden{display:none;}

```

## ● script.js

```

1  /* =====
2      Task 4 – Student Grade Calculator
3      Demonstrates: variables & data types, literals, operators,
4      selection (if/else) and iteration (for loop) statements,
5      and a user-defined function.
6      ===== */
7

```

```

8 // Number of subjects the form collects – a numeric literal used
9 // wherever the subject count is needed, kept as one named constant.
10 const SUBJECT_COUNT = 5;
11
12 /**
13  * getGrade(average)
14  * A user-defined function: takes the numeric average (a Number)
15  * and returns the letter grade (a String) using selection
16  * statements (if / else if).
17  */
18 function getGrade(average) {
19   if (average >= 90) {
20     return "A+";
21   } else if (average >= 80) {
22     return "A";
23   } else if (average >= 70) {
24     return "B";
25   } else if (average >= 60) {
26     return "C";
27   } else if (average >= 40) {
28     return "D";
29   } else {
30     return "F";
31   }
32 }
33
34 /**
35  * calculateResult(marks)
36  * A user-defined function: accepts an array of five marks and
37  * returns an object holding total, average, grade and pass/fail
38  * status. This is where the arithmetic operators and the loop
39  * (iteration statement) are demonstrated.
40  */
41 function calculateResult(marks) {
42   let total = 0; // variable, initialised with a numeric literal
43
44   // Iteration statement: a for loop summing every subject's marks
45   for (let i = 0; i < marks.length; i++) {
46     total = total + marks[i]; // arithmetic operator: addition
47   }
48
49   const average = total / SUBJECT_COUNT; // arithmetic operator: division
50   const grade = getGrade(average);
51
52   // A subject below 35 is treated as a fail in that subject –
53   // the logical operator (some()) checks this across the array.
54   const hasFailingSubject = marks.some(function (mark) {
55     return mark < 35; // relational operator: less-than
56   });
57
58   // Selection statement + logical AND: overall status depends on
59   // both the average and whether any single subject failed.
60   const status = (average >= 40 && !hasFailingSubject) ? "Pass" : "Fail";
61
62   return { total, average, grade, status };
63 }
64
65 // ----- Wiring the function up to the form -----

```

```

66
67 const form = document.getElementById("gradeForm");
68 const resultPanel = document.getElementById("result");
69 const errorMsg = document.getElementById("errorMsg");
70
71 form.addEventListener("submit", function (event) {
72     event.preventDefault(); // stop the page from reloading
73
74     const marks = [];
75     let allValid = true;
76
77     // Iteration statement: read and validate all five subject inputs
78     for (let i = 1; i <= SUBJECT_COUNT; i++) {
79         const input = document.getElementById("s" + i);
80         const value = Number(input.value); // data type conversion: string -> number
81
82         // Selection statement: reject anything outside 0-100
83         if (isNaN(value) || value < 0 || value > 100) {
84             allValid = false;
85         }
86         marks.push(value);
87     }
88
89     if (!allValid) {
90         errorMsg.classList.remove("hidden");
91         resultPanel.classList.add("hidden");
92         return;
93     }
94
95     errorMsg.classList.add("hidden");
96
97     const result = calculateResult(marks);
98
99     document.getElementById("rTotal").textContent = result.total;
100    document.getElementById("rAverage").textContent = result.average.toFixed(2) + "%";
101
102    document.getElementById("rGrade").textContent = result.grade;
103
104    1
105    2
106
107    const statusEl = document.getElementById("rStatus");
108
109    3
110    statusEl.textContent = result.status;
111
112    4
113    statusEl.className = "status " + (result.status === "Pass" ? "pass" : "fail");
114
115    5
116    6
117    resultPanel.classList.remove("hidden");
118
119    7
120    });
121
122    8

```

# Task 5 — Supermarket Billing System

Folder: Assignment-WebTech-2026-08-18/Task5-Billing-System/

● index.html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Supermarket Billing System</title>
7  <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10
11  <main class="bill-wrap">
12    <h1>Supermarket Bill Generator</h1>
13    <p class="subtitle">Enter quantity and unit price for up to three products. A 10% discount
14    applies automatically once the bill crosses ₹2000.</p>
15
16    <form id="billForm">
17      <table class="product-table">
18        <thead>
19          <tr>
20            <th>Product</th>
21            <th>Quantity</th>
22            <th>Unit Price (₹)</th>
23          </tr>
24        </thead>
25        <tbody>
26          <tr>
27            <td><label for="name1">Product 1</label></td>
28            <td><input type="number" id="qty1" min="0" value="0" required></td>
29            <td><input type="number" id="price1" min="0" step="0.01" value="0" required></td>
30          </tr>
31          <tr>
32            <td><label for="name2">Product 2</label></td>
33            <td><input type="number" id="qty2" min="0" value="0" required></td>
34            <td><input type="number" id="price2" min="0" step="0.01" value="0" required></td>
35          </tr>
36          <tr>
37            <td><label for="name3">Product 3</label></td>
38            <td><input type="number" id="qty3" min="0" value="0" required></td>
39            <td><input type="number" id="price3" min="0" step="0.01" value="0" required></td>
40          </tr>
41        </tbody>
42      </table>
43
44      <button type="submit">Generate Bill</button>
45    </form>
46
47    <section id="bill" class="bill-panel hidden">
48      <h2>Bill Summary</h2>
49      <div class="bill-row"><span>Subtotal</span><strong id="bSubtotal">₹0.00</strong></div>
```

```

49     <div class="bill-row" id="discountRow"><span>Discount (10%)</span><strong
id="bDiscount">₹0.00</strong></div>
50     <div class="bill-row total"><span>Payable Amount</span><strong id="bTotal">₹0.00</strong></div>
51     <p id="discountNote" class="discount-note hidden">🎉 Discount applied – your order crossed
₹2000!</p>
52     </section>
53     </main>
54
55     <script src="script.js"></script>
56 </body>
57 </html>

```

## ● style.css

```

1  /* Task 5 – Supermarket Billing System styling */
2
3  :root{
4      --ink:#20242b;
5      --paper:#f2f4f8;
6      --panel:#ffffff;
7      --accent:#e08a2c;
8      --line:#dbe0ea;
9      --good:#1f8a4c;
10 }
11
12 *{box-sizing:border-box;}
13 body{
14     margin:0;
15     min-height:100vh;
16     display:flex;
17     align-items:center;
18     justify-content:center;
19     background:var(--paper);
20     font-family:'Segoe UI', system-ui, sans-serif;
21     color:var(--ink);
22     padding:30px 16px;
23 }
24
25 .bill-wrap{
26     width:100%;
27     max-width:480px;
28     background:var(--panel);
29     border:1px solid var(--line);
30     border-radius:12px;
31     padding:32px 30px;
32     box-shadow:0 20px 40px -28px rgba(20,25,40,0.35);
33 }
34
35 h1{margin:0 0 6px;font-size:24px;}
36 .subtitle{margin:0 0 22px;font-size:14px;color:#5a6072;}
37
38 .product-table{width:100%;border-collapse:collapse;margin-bottom:20px;}
39 .product-table th{
40     text-align:left;font-size:12px;text-transform:uppercase;letter-spacing:0.04em;
41     color:#5a6072;padding-bottom:8px;border-bottom:1px solid var(--line);
42 }
43 .product-table td{padding:8px 6px 8px 0;}
44 .product-table td:first-child{font-size:14px;font-weight:600;width:30%;}

```

```

45 .product-table input{
46   width:100%;padding:9px 10px;
47   border:1.5px solid var(--line);border-radius:7px;
48   font-size:14px;font-family:inherit;
49 }
50 .product-table input:focus{outline:none;border-color:var(--accent);}
51
52 button[type="submit"]{
53   width:100%;padding:13px;
54   background:var(--accent);color:#fff;
55   border:none;border-radius:8px;
56   font-size:15px;font-weight:700;cursor:pointer;
57 }
58 button[type="submit"]:hover{background:#c47420;}
59
60 .bill-panel{
61   margin-top:24px;padding:20px 22px;
62   background:var(--paper);border-radius:10px;
63   border:1px solid var(--line);
64 }
65 .bill-panel h2{margin:0 0 14px;font-size:16px;}
66 .bill-row{
67   display:flex;justify-content:space-between;
68   padding:8px 0;font-size:14px;
69   border-bottom:1px dashed var(--line);
70 }
71 .bill-row.total{
72   border-bottom:none;border-top:1.5px solid var(--ink);
73   margin-top:4px;padding-top:12px;
74   font-size:16px;font-weight:700;
75 }
76 .discount-note{
77   margin:14px 0 0;padding:10px 12px;
78   background:#e6f5ec;color:var(--good);
79   border:1px solid #a9dfbf;border-radius:8px;
80   font-size:13px;
81 }
82 .hidden{display:none;}

```

## ● script.js

```

1  /* =====
2   Task 5 – Supermarket Billing System
3   Demonstrates: variables & data types, literals, operators,
4   statements (selection), and a user-defined function.
5   ===== */
6
7  // Literals: the discount rate and the threshold that triggers it
8  const DISCOUNT_RATE = 0.10;      // numeric literal – 10%
9  const DISCOUNT_THRESHOLD = 2000; // numeric literal – ₹2000
10
11 /**
12  * calculateBill(products)
13  * A user-defined function: accepts an array of {quantity, price}
14  * objects, sums them into a subtotal, applies the discount rule
15  * with a selection (if) statement, and returns the breakdown.
16  */
17 function calculateBill(products) {

```

```

18 let subtotal = 0; // variable, Number data type
19
20 // Iteration statement: total up every product line
21 for (let i = 0; i < products.length; i++) {
22     const lineTotal = products[i].quantity * products[i].price; // arithmetic operator: *
23     subtotal = subtotal + lineTotal; // arithmetic operator: +
24 }
25
26 let discount = 0;
27 let discountApplied = false;
28
29 // Selection statement: only discount orders over the threshold
30 if (subtotal > DISCOUNT_THRESHOLD) {
31     discount = subtotal * DISCOUNT_RATE;
32     discountApplied = true;
33 }
34
35 const payable = subtotal - discount; // arithmetic operator: subtraction
36
37 return { subtotal, discount, payable, discountApplied };
38 }
39
40 // ----- Wiring the function up to the form -----
41
42 const form = document.getElementById("billForm");
43
44 form.addEventListener("submit", function (event) {
45     event.preventDefault();
46
47     const products = [];
48
49     // Iteration statement: read quantity + price for all 3 products
50     for (let i = 1; i <= 3; i++) {
51         const quantity = Number(document.getElementById("qty" + i).value);
52         const price = Number(document.getElementById("price" + i).value);
53         products.push({ quantity: quantity, price: price });
54     }
55
56     const bill = calculateBill(products);
57
58     document.getElementById("bSubtotal").textContent = "₹" + bill.subtotal.toFixed(2);
59     document.getElementById("bDiscount").textContent = "- ₹" + bill.discount.toFixed(2);
60     document.getElementById("bTotal").textContent = "₹" + bill.payable.toFixed(2);
61
62     // Selection statement: only show the discount row/note when it applies
63     const discountRow = document.getElementById("discountRow");
64     const discountNote = document.getElementById("discountNote");
65     if (bill.discountApplied) {
66         discountRow.classList.remove("hidden");
67         discountNote.classList.remove("hidden");
68     } else {
69         discountRow.classList.add("hidden");
70         discountNote.classList.add("hidden");
71     }
72
73     document.getElementById("bill").classList.remove("hidden");
74 });

```

— End of source listing —